Chapter 2 — OOP Fundamentals (Study Edition)

> **Study Edition.** Every topic of the full chapter, without the padding. Full programs and outputs are in the Full Edition (`README.md`).
>
> **You will learn:** the class basics, the four pillars on one bank example, and the Java OOP tools LLD relies on (sections 2.2–2.17 map one-to-one to the Full Edition).

Run all examples: `cd chapters/02-oop-fundamentals/code && javac --release 21 -d /tmp/lld_02 $(find . -name '*.java') && java -cp /tmp/lld_02 lld.ch02.<package>.Main`

2.1 What object-oriented programming is

What it is

OOP bundles data with the only code allowed to modify it (an **object**). Procedural code keeps data apart, so any function can change it. Analogy: cash on a table vs a vault with a teller (the methods) who checks every request.

The problem it solves

A rule like "balance never negative" lives **once**, next to its data, instead of in every caller; designs grow by **adding classes**.

Why this matters for the interview

LLD tests **splitting a problem into objects, giving each responsibilities and data, and justifying it**.

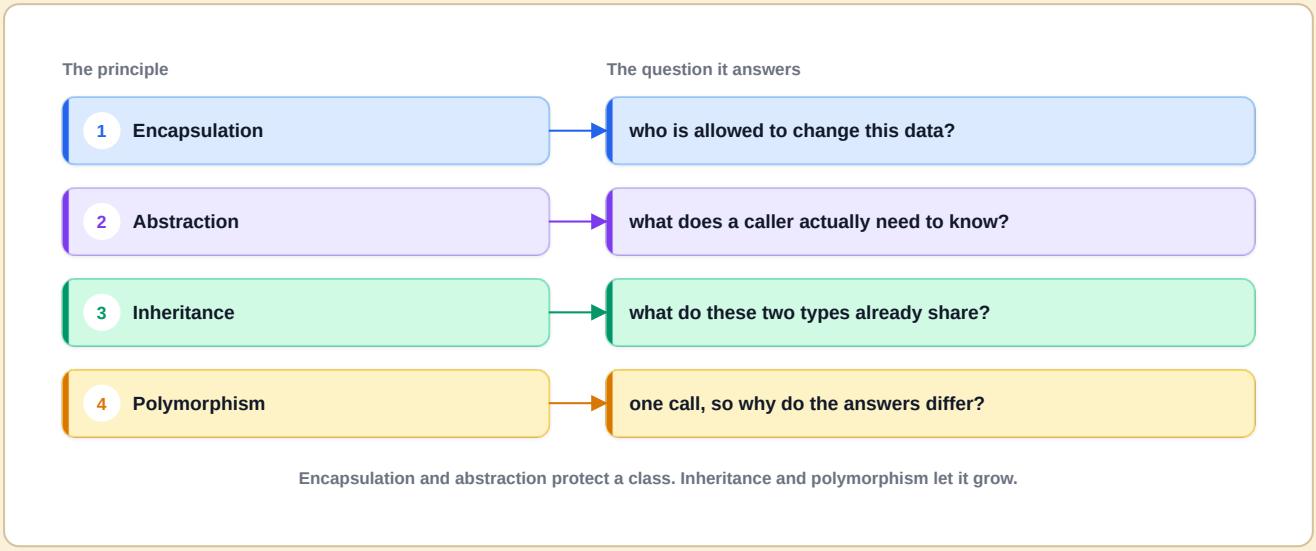


Figure — The four OOP principles and the question each one answers (course diagram)

Pillar	Question it answers
Encapsulation	Who may change this data?
Abstraction	What does a caller need to know?
Inheritance	What do these types share?
Polymorphism	One call, why do answers differ?

The first two guard one class; the last two let the design grow with new classes.

2.2 Classes, objects, constructors, `this`, `static` and `final`

What it is

Class = blueprint; **object** = one instance made with `new`. **Constructor** = named like the class, **no return type**, makes the object **valid**. `this` = current object. `static` = one per class, shared (else per-object **instance** members). `final` = variable set once / method not overridable / class not extendable.

How it works

1. `new` sets fields to defaults (0, `null`, `false`), then runs the constructor.
2. **Order:** static initialisers run **once** at first use; per `new`: parent constructor (`super(...)`) → instance initialisers in source order → constructor body.
3. No constructor written → compiler adds a public no-arg **default constructor**; write any → it doesn't.
4. **Chain** overloads to one canonical constructor with `this(...)` so validation lives once; in Java 21 it must be the **first statement** (Java 25, JEP 513, relaxes this).
5. `this`: disambiguate (`this.id = id`), pass self, chain, `return this` for **fluent** calls.
6. `static` methods have **no** `this`: utilities and factories.
7. Variables hold **references**: `Account alias = a;` is the **same** object.

Java example

```java
// File: code/lld/ch02/classes/Account.java
public class Account {
    // static = belongs to the CLASS, shared by all objects
    private static int openedCount = 0;
    // ... MIN_OPENING_CENTS constant, final id/owner, balanceCents

    public Account(String id, String owner, long openingCents) {
        if (openingCents < MIN_OPENING_CENTS) throw new IllegalArgumentException("negative opening balance");
        this.id = id;                 // 'this.id' is the field, 'id' is the parameter
        this.owner = owner;
        this.balanceCents = openingCents;
        openedCount++;                // shared counter
    }

    public Account(String id, String owner) {
        this(id, owner, 0);
    }

    public Account deposit(long cents) {   // returns 'this' so calls can be chained
        if (cents <= 0) throw new IllegalArgumentException("deposit must be positive");
        this.balanceCents += cents;
        return this;
    }
    // ... accessors, static openedCount(), toString
}
```

Full runnable code: code/lld/ch02/classes/ (see the Full Edition).

A refused `new Account("X", "Bad", -5)` leaves the counter unchanged: **validate first, then change state**.

Trade-offs

Static members are shared but are global state (hard to test, thread-risky). Many constructor overloads confuse past 3–4 (use a Builder or static factories). `final` fields are safer but must be set in every constructor.

Common mistakes

Per-object data in a `static` field; heavy work or overridable calls in constructors; `id = id;`; thinking `=` copies an object.

Real-world examples

`Math.max` (static); `StringBuilder.append` returns `this`; `String` is `final`.

In the interview

> **30-second answer:** "A class is the blueprint, an object an instance with its own state. Constructors create valid objects; I chain overloads with `this(...)`. Static members are shared, so I keep them for constants, utilities and factories, never mutable state."

Self-check

- **Q:** Copies of a `static` field for 1,000 objects? **A:** One.
- **Q:** Parent constructor or child initialisers first? **A:** Parent constructor.

TL;DR

Class = blueprint, object = instance; chain with `this(...)`; `static` = per class; variables hold references.

2.3 Encapsulation

What it is

Hide data (**private**) and allow change **only through guarded methods**. Analogy: a child-proof cap.

The problem it solves

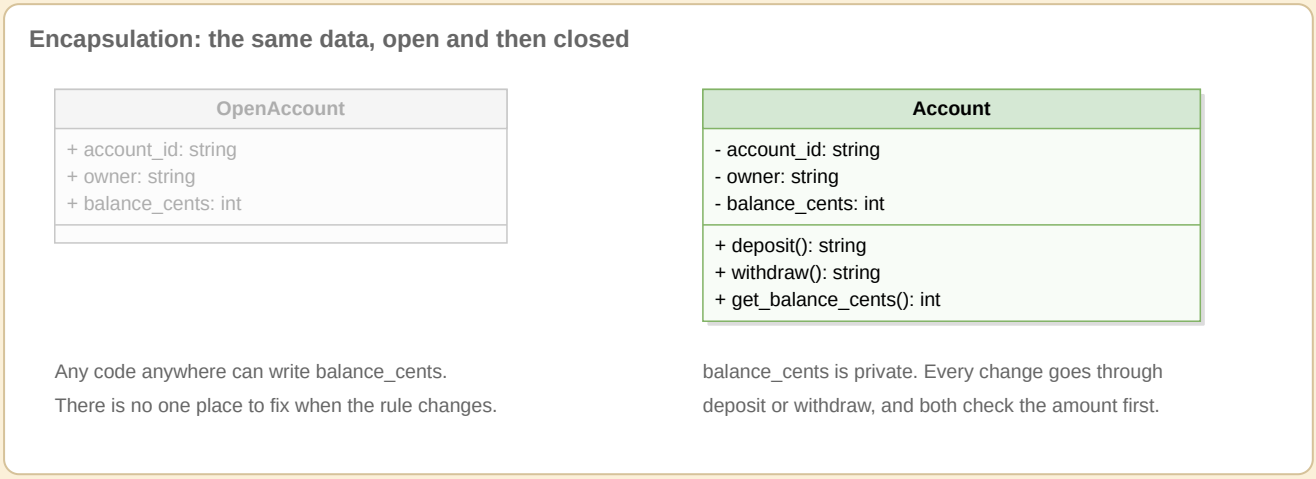


Figure — The same account data, first open (public fields anyone can write) and then closed (private fields changed only through deposit and withdraw) (course diagram)

With a private balance, **every route to the data passes a method that can say no**; public fields leave no central place for the rule.

How it works

Fields `private` (and `final`); expose **behaviour** (`withdraw`), not setters; validate there, **once**; return copies/unmodifiable views of collections; decide **who may modify each field** and say it.

Java example

`Account` in §2.6: private balance, `final deposit`/`withdraw`.

Trade-offs · When not to use it (strictly)

Rules in one place and freely changeable internals, for a little more code. Rule-free data carriers (DTOs, events): use a **record**.

Common mistakes

Getter + setter for every field (anemic, private in name only); returning the live internal list.

Real-world examples

`String`, `ArrayList`; `LocalDate.of(2026, 2, 30)` throws.

In the interview

> **30-second answer:** "State hides behind methods that hold every rule about it. If only this class may change a field, it's private and changed through `withdraw`-style methods, not setters." A rule-free setter is a public field in disguise.

Self-check

- **Q:** Where does "no negative balance" live? **A:** In `Account.withdraw`, once.

TL;DR

Private data + rule-enforcing methods; behaviour, not setters.

2.4 Abstraction

What it is

Give callers **only the operations they need**, hide **how**. Encapsulation hides **state**; abstraction hides **decisions**. Analogy: pedals work the same for petrol or electric.

The problem it solves

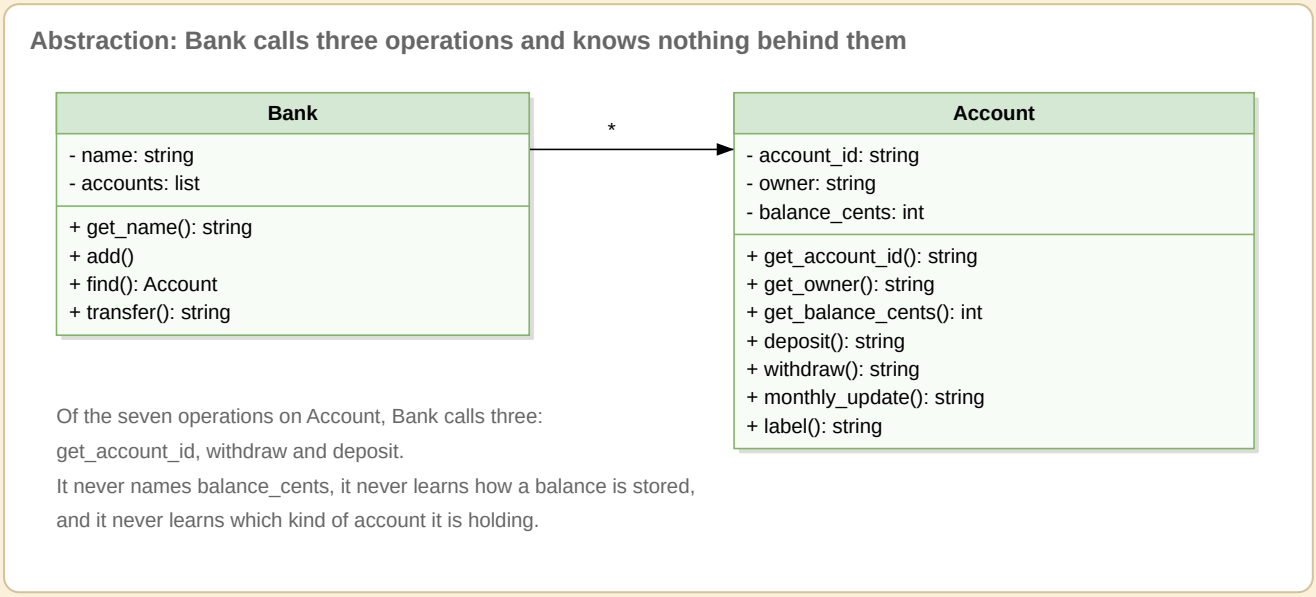


Figure — Of Account's seven public operations, Bank relies on three and sees nothing of what lies behind them (course diagram)

`Bank.transfer` uses only `id()`, `withdraw()`, `deposit()` (our Java `Account` has six public operations, the diagram seven). It doesn't know about cents or subclasses, so storage could move to a database untouched.

How it works

Smallest set of operations, named by **what** they do, on an interface/public surface; callers depend only on it.

Java example

`Bank.transfer` in §2.6.

Trade-offs

Simpler callers, enables polymorphism; a wrong abstraction is costly, too many layers confuse.

Common mistakes

Leaky abstractions (`getHashMapOfSeats()`); abstracting before two concrete cases.

Real-world examples

`List`, JDBC `Connection`, `InputStream`.

In the interview

> **30-second answer:** "Expose only what callers need; whenever 'the caller doesn't need to know that' is true, it moves behind the interface." Encapsulation protects **data**; abstraction hides **complexity**.

Self-check

- **Q:** What does `transfer` need from `Account`? **A:** `id()`, `withdraw()`, `deposit()`.

TL;DR

Minimum surface, hidden how.

2.5 Inheritance

What it is

A **child** gets everything its **parent** has, then **adds** or **overrides**: `class SavingsAccount extends Account` (a savings account **is an** account).

The problem it solves

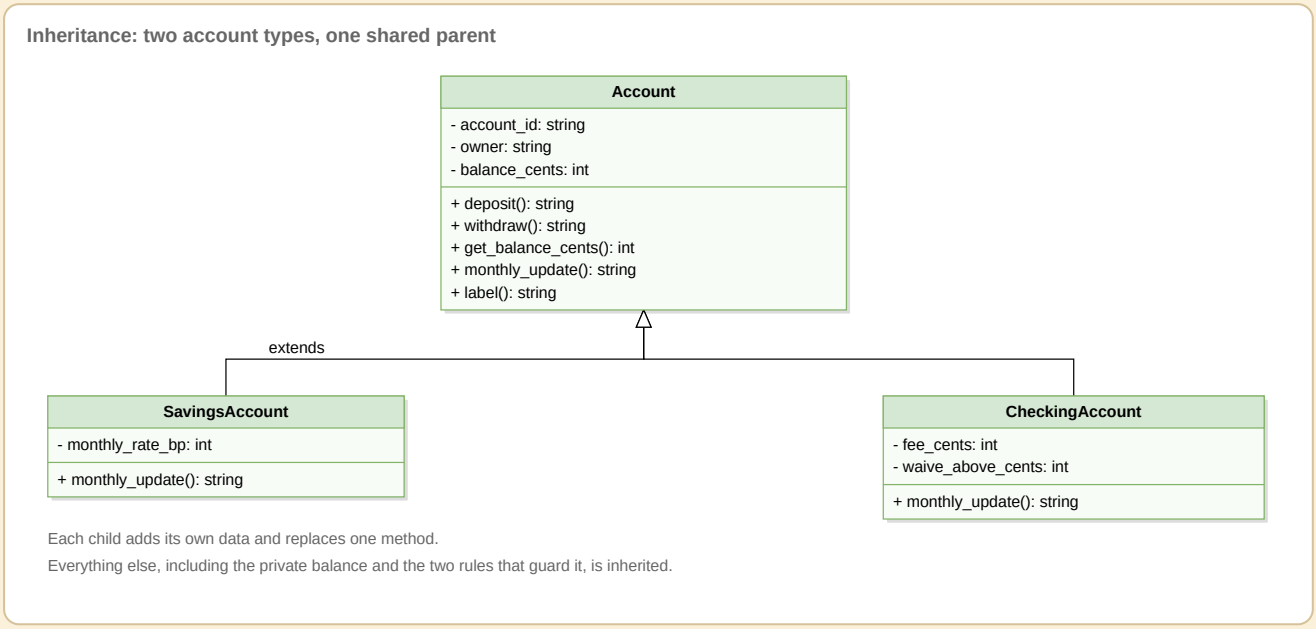


Figure — SavingsAccount and CheckingAccount extend Account; each adds its own data and replaces one method (course diagram)

Both children share the parent's balance and rules, add 1–2 fields and **override one method**. If is-a were false, the difference would be a **field**.

How it works

1. **One** parent class, many interfaces. **Constructors are not inherited**; private fields exist but are reached via parent methods.
2. Child constructor calls `super(...)` first (implicit `super()` fails if the parent lacks a no-arg constructor). `super.method()` reuses the parent's code.
3. **Overriding rules:**

Rule	Detail
Signature	Same name and parameter types
Return type	Same or a **subtype** (*covariant*); primitives exact
Access	Same or **wider** only
Checked exceptions	Fewer/narrower only
Cannot override	`final`; `static` (**hidden**); `private`
Annotation	Always `@Override`

4. Every class extends `Object` (`toString`, `equals`, `hashCode`, `getClass`). The **is-a test** must hold always.

Java example

`SavingsAccount`/`CheckingAccount` (§2.6); `Circle` (§2.7).

Trade-offs

Reuse and polymorphism for true families; but **tight coupling** to parent internals (fragile base class), compile-time fixed, single parent, hard past 2–3 levels.

When not to use it

Code reuse without is-a (use composition); a child that must **disable** a parent method (breaks **Liskov**, Chapter 4); data-only differences (field/enum).

Common mistakes

`Square extends Rectangle` with independent setters; missing `@Override` (accidental overload); overridable calls in parent constructors.

Real-world examples

Good: `FileInputStream extends InputStream`. JDK mistakes: `Stack extends Vector`, `Properties extends Hashtable`.

In the interview

> **30-second answer:** "Inheritance is right only when 'child is a kind of parent' always holds. I use `@Override`, and if a child must disable a parent method I switch to composition, since that breaks Liskov."

- *"Override a static method?"* — No, it's **hidden** (chosen by compile-time type).

Self-check

- **Q:** Override from `public` to `protected`? **A:** No.

TL;DR

True is-a only; `super(...)` first, `@Override` always.

2.6 Polymorphism (and the full bank example)

What it is

One call, different behaviour per receiving object. Code holds the **parent type**; the actual object decides. Analogy: one "start" button on a car, washer, laptop.

The problem it solves

Removes `if (a instanceof SavingsAccount) ... else if ...` chains that every new type must edit.

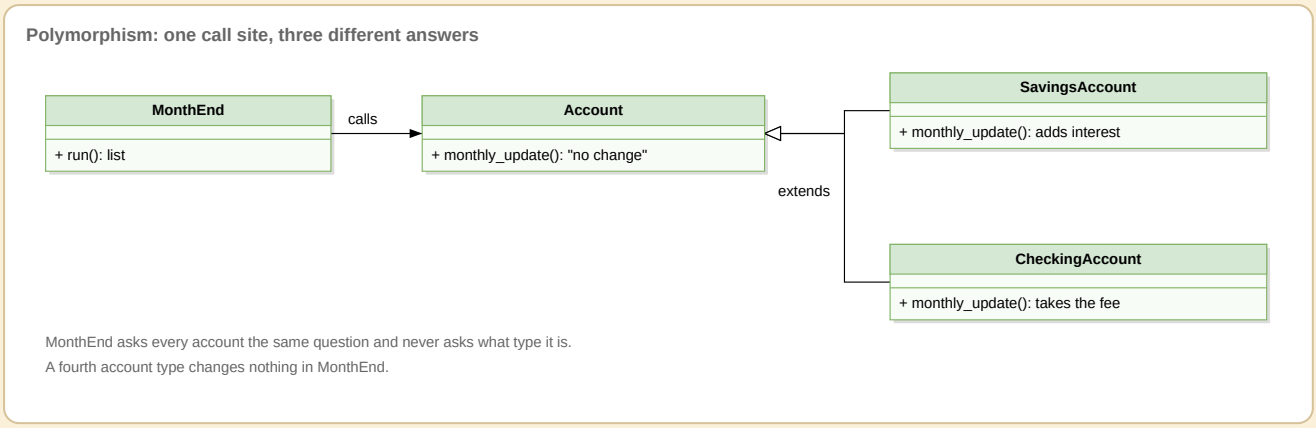


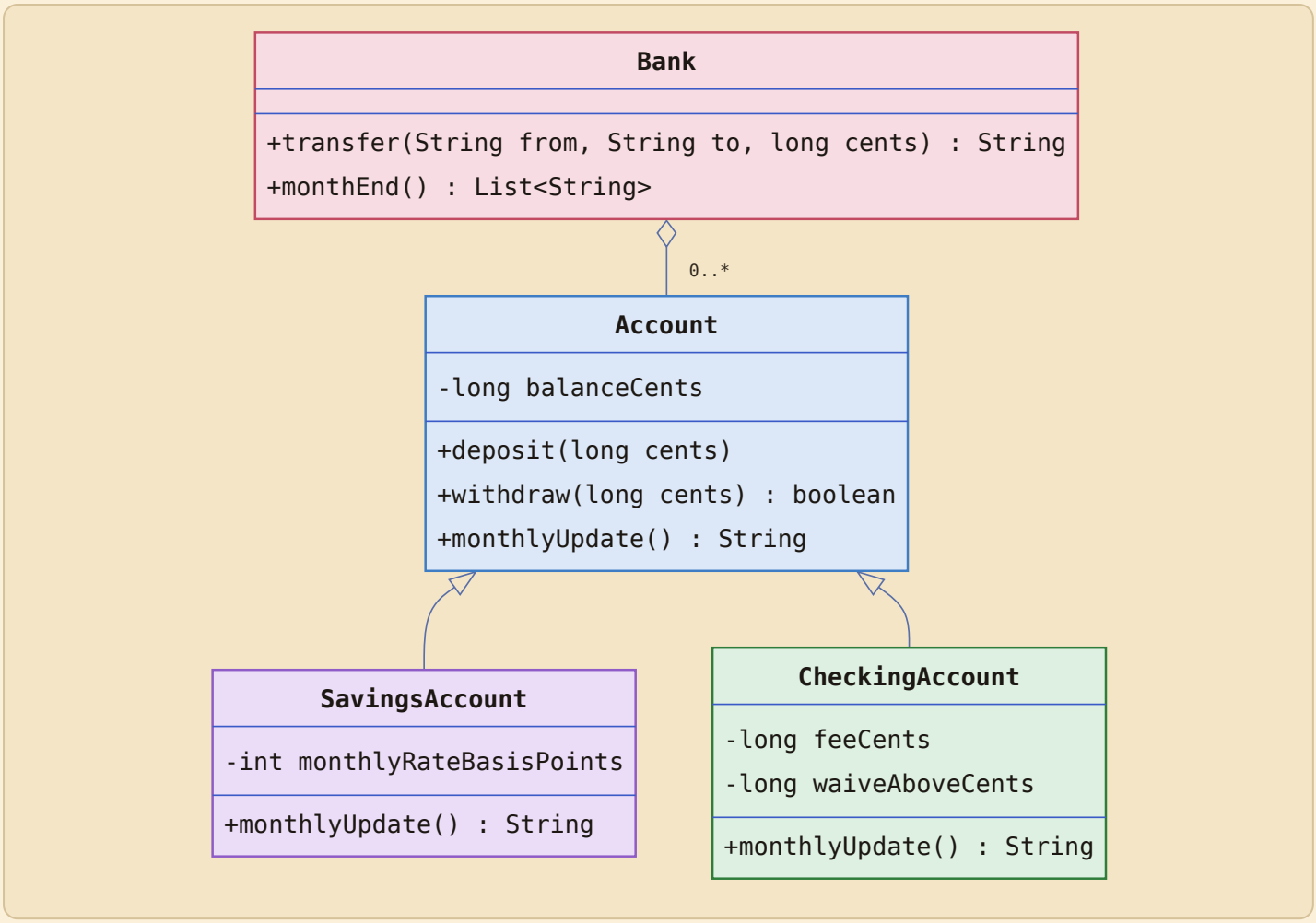
Figure — MonthEnd calls monthlyUpdate on every account and never asks which type it holds (course diagram)

The month-end loop has **no type** `if`; a fourth account type needs no edit.

How it works

Declare on the parent/interface → children override → callers hold the parent type → the JVM runs the **actual class's** version: **dynamic dispatch** (late binding).

Java example: the bank (encapsulation + abstraction + inheritance + polymorphism)




```java
// File: code/lld/ch02/bank/Account.java
public class Account {
    private long balanceCents;                    // hidden: nobody outside can set it to -5
    // ... id, owner, constructor

    public final boolean withdraw(long cents) {
        if (cents <= 0 || cents > balanceCents) return false;   // the one place the rule lives
        balanceCents -= cents;
        return true;
    }
    // ... final deposit() and accessors

    public String monthlyUpdate() { return "no change"; }
}
```

```java
// File: code/lld/ch02/bank/SavingsAccount.java
    @Override
    public String monthlyUpdate() {
        long interest = balanceCents() * monthlyRateBasisPoints / 10_000;
        if (interest > 0) deposit(interest);
        return "interest " + Money.format(interest);
    }
```

```java
// File: code/lld/ch02/bank/Bank.java
    public List<String> monthEnd() {
        List<String> lines = new ArrayList<>();
        for (Account a : accounts) {
            String what = a.monthlyUpdate();                // dynamic dispatch picks the override
            lines.add(a.id() + " (" + a.owner() + "): " + what + ", balance " + Money.format(a.balanceCents()));
        }
        return lines;
    }
```

Full runnable code: code/lld/ch02/bank/ (see the Full Edition).

```
S-1 rejected, not enough money
month end:
  S-1 (Ada): interest $1.00, balance $401.00
  C-1 (Kai): fee $2.50, balance $177.50
  P-1 (Ravi): no change, balance $30.00
```

- The $900 transfer is refused inside `withdraw` (encapsulation); `CheckingAccount` waives its fee above a threshold; one call gives three results with no type `if` (polymorphism).
- **Why `long` cents:** exact, unlike `double`, but integer division **truncates** fractions of a cent. For interest/tax use `BigDecimal` with a `RoundingMode` (§2.13). 1 basis point = 0.01%.

How the four fit together

Encapsulation + abstraction decide **what is hidden**; inheritance + polymorphism make **new requirements into added classes**.

Trade-offs

Kills type switches and underpins most patterns; but behaviour spreads across classes (virtual-call cost rarely matters).

Common mistakes

Type `switch`/`if` chains (**pointing it out earns more credit than refactoring**); `instanceof` chains as the main dispatch.

Real-world examples

`List.add` on `ArrayList` vs `LinkedList`; `Comparator.compare`.

In the interview

30-second answer: "I code against the parent type or interface, each subtype overrides, and the JVM picks at run time, so a new type is a new class, not an edit." In a parking lot: pricing, payments, spot selection.

Self-check

- **Q:** `Account a = new SavingsAccount(...)`: whose `monthlyUpdate`? **A:** `SavingsAccount`'s, at run time.

TL;DR

Hold the parent type, let the object decide; type switches signal missing polymorphism.

2.7 Compile-time vs runtime polymorphism: overloading vs overriding

What it is

	Overloading (compile-time)	Overriding (runtime)
What	Same name, **different parameters**	Same signature in a **subclass**
Chosen by	**Compiler**, from **declared** argument types	**JVM**, from the receiver's **actual class**
Return type	May differ (not alone)	Same or covariant
Inheritance?	No	Yes

How it works: overload resolution

Phases, first match wins: (1) exact or **widening** (`int` → `long` → `double`, `Circle` → `Shape`); (2) **boxing**; (3) **varargs**. Ties → **most specific**, else ambiguous error. **Not polymorphic:** fields (**hidden**), `static` and `private` methods.

Java example

```
// File: code/lld/ch02/polymorphism/Calculator.java
    public String add(int a, int b)       { return "add(int,int) = " + (a + b); }
    public String add(long a, long b)     { return "add(long,long) = " + (a + b); }
    // ... add(double,double), add(int...)
    public String greet(Shape s)  { return "greet(Shape)"; }
    public String greet(Circle c) { return "greet(Circle)"; }
```

```
// File: code/lld/ch02/polymorphism/Circle.java
public class Circle extends Shape {
    public String kind = "circle";                              // HIDES Shape.kind, does not override it
    // ...
    @Override public double area() { return Math.PI * r * r; }  // OVERRIDE: same signature

    @Override public Circle copy() { return new Circle(r); }    // covariant return type is allowed
    // ... static describe() (hidden), who() calls super.who()
}
```

Full runnable code: code/lld/ch02/polymorphism/ (see the Full Edition).

With `Shape s = new Circle(1.0)`: `calc.add(2L, 3)` → `add(long,long)` (widening); `s.area()` → 3.14 (dispatch); `calc.greet(s)` → **`greet(Shape)`** (overloads use the declared type: a favourite question); `s.kind` → `shape` (**never hide fields**); `s.copy()` is a `Circle`. `s instanceof Circle c` is pattern matching (Java 16+).

Common mistakes

Expecting overloads to follow runtime type (needs double dispatch: Visitor, Chapter 11); `equals(Point p)` overload.

In the interview

> **30-second answer:** "Overloading is compile-time, chosen by declared types; overriding is runtime, dispatched on the actual object. Fields and statics aren't polymorphic."

Self-check

- **Q:** Differ only in return type? **A:** Compile error. **Q:** Varargs phase? **A:** Last.

TL;DR

Overloading = compiler, declared types; overriding = JVM, actual object.

2.8 Interfaces (default, static, private methods, functional interfaces)

What it is

A **contract**: methods a class promises. Analogy: a socket standard. Abstract methods are `public abstract`, fields `public static final`; a class **implements many**. **Default** (Java 8) bodies let interfaces evolve (`Collection.stream()`); **static** (Java 8) belong to the interface, not inherited; **private** (Java 9) help defaults. A **functional interface** has **exactly one abstract method** (Object/default/static don't count) → a **lambda** can implement it; `@FunctionalInterface` enforces it.

The problem it solves

Callers depend on **what**, not a class: add implementations without edits, test with fakes, multiple inheritance of **type** (not state).

How it works

Smallest method set; implement via classes or lambdas; callers use the interface type. Same default in two interfaces → class **must** override (may call `A.super.m()`); a superclass method **wins** over defaults.

Java example

```java
// File: code/lld/ch02/interfaces/Notifier.java
public interface Notifier {
    String send(String to, String message);        // abstract method: every implementer must provide it

    default String sendAll(java.util.List<String> recipients, String message) {
        StringBuilder sb = new StringBuilder();
        for (String r : recipients) sb.append(send(r, clean(message))).append('\n');
        return sb.toString().stripTrailing();
    }

    static Notifier console() { return (to, msg) -> "[console] " + to + ": " + msg; }

    private String clean(String message) { return message.strip(); }
}
```

Full runnable code: code/lld/ch02/interfaces/ (see the Full Edition).

`EmailNotifier` and a lambda both implement `Notifier`; `DiscountRule` is an `@FunctionalInterface` with a default `then` combinator. JDK ones: `Function`, `Supplier`, `Consumer`, `Predicate`.

Trade-offs

Loose coupling, testability, evolvable; but an extra type, no state, and overused defaults become mini base classes.

When not to use it

One implementation, no realistic second, no test double.

Common mistakes

Fat interfaces (ISP); constant-only interfaces.

Real-world examples

`Comparable`, `Runnable`, `AutoCloseable`, Spring `Repository`.

In the interview

30-second answer: "Services depend on interfaces like `PaymentGateway` so implementations can be added and faked. Interfaces have default and static methods; one abstract method makes them lambda-friendly." Defaults exist to add methods like `stream` without breaking implementers.

Self-check

- **Q:** Is `Comparator` functional? **A:** Yes; only `compare` is abstract.

TL;DR

Depend on contracts; defaults evolve, statics are factories, one abstract method = lambda.

2.9 Abstract classes vs interfaces

What it is

An **abstract class** can't be instantiated; it may hold fields, constructors, finished and **abstract** methods. Analogy: a partly pre-printed form.

How it works

```java
// File: code/lld/ch02/interfaces/Report.java
public abstract class Report {
    private final String title;                    // state: interfaces cannot have instance fields

    protected Report(String title) { this.title = title; }

    public final String render() {                 // final: subclasses cannot change the skeleton
        return header() + "\n" + body() + "\n-- end of " + title + " --";
    }

    protected String header() { return "== " + title + " =="; }   // concrete, overridable
    protected abstract String body();                              // must be supplied
}
```

`SalesReport` supplies only `body()`: **Template Method** (Chapter 11).

	Abstract class	**Interface**
State / constructors	Yes	Constants only / no
Bodies	Any	`default`, `static`, `private`
Per class	**One**	**Many**
Expresses	"is a"	"can do"
Use	Shared state + partial implementation	Contracts, roles, strategies, test seams

Rule of thumb

Interface for the contract; abstract class beneath when implementations share state or an algorithm (`List` + `AbstractList` + `ArrayList`).

Common mistakes

Abstract class to share one helper; everything abstract "for flexibility".

In the interview

30-second answer: "Interfaces define capabilities and allow many; abstract classes share state and partial implementation, one parent. Interface first; abstract base for real shared state or a template."

Self-check

- **Q:** Abstract class without abstract methods? **A:** Allowed. **Q:** Interface constructor? **A:** No.

TL;DR

Interface = contract, many, stateless; abstract class = partial implementation, one.

2.10 Enums with fields and behaviour

What it is

A **fixed set of named instances**; in Java a full class with **fields, constructors, methods, interfaces**, even per-constant bodies.

The problem it solves

String/int constants allow typos and carry no behaviour. Enums are **type-safe** and hold each value's data and rules, removing switches: LLD's home for "**different data, not behaviour**".

How it works

Implicitly extend `Enum`; private constructors; `values()`, `valueOf` (throws on unknown), `name()`, `ordinal()` (**never persist**); compare with `==`; per-constant bodies make tiny state machines; `EnumMap`/`EnumSet`; exhaustive `switch` needs no `default`; a one-element enum is the simplest safe **Singleton**.

Java example

```java
// File: code/lld/ch02/enums/OrderStatus.java
public enum OrderStatus {
    PLACED    { @Override Set<OrderStatus> next() { return Set.of(PAID, CANCELLED); } },
    PAID      { @Override Set<OrderStatus> next() { return Set.of(SHIPPED, CANCELLED); } },
    SHIPPED   { @Override Set<OrderStatus> next() { return Set.of(DELIVERED); } },
    DELIVERED { @Override Set<OrderStatus> next() { return Set.of(); } },
    CANCELLED { @Override Set<OrderStatus> next() { return Set.of(); } };

    abstract Set<OrderStatus> next();          // constant-specific method bodies

    public boolean canMoveTo(OrderStatus target) { return next().contains(target); }
    // ... isFinal()
}
```

Full runnable code: code/lld/ch02/enums/ (see the Full Edition).

`Coin` carries `cents` as a field (`PENNY(1) ... QUARTER(25)`), so summing needs no switch. `OrderStatus` refuses `PLACED -> SHIPPED`: the "**enum + guarded method**" alternative to the State pattern.

Trade-offs

Type-safe, data + behaviour; but closed (recompile to add), and large per-constant logic belongs in State/Strategy classes.

When not to use it

Values created at run time, or needing big independent behaviour.

Common mistakes

Persisting `ordinal()`; switches over the enum instead of methods on it.

Real-world examples

`DayOfWeek`, `TimeUnit.SECONDS.toMillis(5)`, `Thread.State`.

In the interview

> **30-second answer:** "Enums model fixed sets like vehicle types and statuses, carrying data like 'which spots fit' or 'which status may follow'." Enum if only data differs; classes if behaviour does.

Self-check

- **Q:** `Coin.valueOf("dime")`? **A:** `IllegalArgumentException` (case-sensitive).

TL;DR

Type-safe fixed sets with data and behaviour; `==`; no persisted ordinals.

2.11 Composition over inheritance

What it is

Hold other objects and **delegate** ("has-a") instead of inheriting ("is-a"); GoF's "favour composition over inheritance". Analogy: a phone **has** an upgradable camera module.

The problem it solves

Inheritance ties the child to the parent's **internals** (**fragile base class**), fixed at compile time. Composition stays independent and **swappable at run time**.

How it works

Put the behaviour behind an interface → hold it in a private field from the constructor → **forward** calls, adding your logic.

Java example

```java
// File: code/lld/ch02/composition/InheritingCountingSet.java
    @Override public boolean add(E e) { attempts++; return super.add(e); }

    @Override public boolean addAll(Collection<? extends E> c) { attempts += c.size(); return super.addAll(c); }
```

```java
// File: code/lld/ch02/composition/CountingSet.java
public class CountingSet<E> {
    private final Set<E> inner;
    // ...
    public boolean addAll(Collection<? extends E> c) {
        attempts += c.size();
        boolean changed = false;
        for (E e : c) changed |= inner.add(e);     // talk to the inner set, not to our own add()
        return changed;
    }
}
```

Full runnable code: code/lld/ch02/composition/ (see the Full Edition).

```
inheritance: added 3, attempts counted = 6
composition: 4 attempts counted = 4, size = 3
```

`HashSet.addAll` calls `add` per element (parent **self-use**, not promised), so the subclass double counts. `CountingSet` works with any `Set`. `Car` **has an** `Engine`, swapped at run time.

Trade-offs

	Inheritance	Composition
Relationship / coupling	is-a / tight	has-a / loose
Run-time change	No	Yes
Boilerplate	Low	Forwarding methods

When not to use it

A genuine is-a class **designed for extension** is fine to extend.

Common mistakes

Extending concrete library classes (`HashMap`); exposing the inner object (`getInner()`).

Real-world examples

`BufferedReader` wraps a `Reader`; Spring DI; `Comparator` in `TreeMap`.

In the interview

> **30-second answer:** "I hold an interface-typed collaborator and delegate: no fragile base class, run-time swapping, and it underlies Strategy, Decorator, State. Inheritance only for true is-a." Example of breakage: the counting set.

Self-check

- **Q:** Why `inner.add`, not `this.add`? **A:** No double counting; relies only on the public contract.

TL;DR

Has-a + delegate beats is-a for reuse.

2.12 Object identity vs equality: the `equals` / `hashCode` contract

What it is

Identity `==`: same object. **Equality** `equals`: same value (default is identity; `String`, `Integer`, records override). Two copies of a book: equal, not identical.

The problem it solves

Hash collections pick a bucket by `hashCode()`, then compare with `equals()`; get either wrong and objects "disappear".

How it works: the contracts

`equals`: reflexive, symmetric, transitive, consistent, `equals(null)` false. `hashCode`: stable, **equal objects → equal hashes**, collisions allowed. **Override both on the same fields.**

Java example

```
// File: code/lld/ch02/equality/Point.java
    @Override
    public boolean equals(Object o) {
        if (this == o) return true;                          // reflexive fast path
        if (!(o instanceof Point p)) return false;           // also handles null
        return x == p.x && y == p.y;
    }

    @Override public int hashCode() { return Objects.hash(x, y); }
```

Full runnable code: code/lld/ch02/equality/ (see the Full Edition).

A `HashSet<Point>` holding two equal points has size 1. `BrokenPoint` (only `equals`) gives `set.contains(equal one): false`; `javac -Xlint` warns `[overrides]`: treat as an error. Compare boxed numbers with `equals`: `Integer` caches only small values (at least −128..127).

Trade-offs / design choice

Value objects (Money, Point): value equality, immutable. **Entities** (User, Order): identity or stable **ID**.

Common mistakes

`equals` without `hashCode`; `equals(Point p)`; **mutable fields in `hashCode`** then mutating a key; mixing `getClass()` / `instanceof`.

Real-world examples

`String`, `LocalDate`, `List.equals`; `new BigDecimal("1.0").equals(new BigDecimal("1.00"))` is **false** (scale; use `compareTo`).

In the interview

30-second answer: "`==` is identity, `equals` value. I override `hashCode` with `equals` on the same fields, since hash collections locate the bucket by hash. Records for values, IDs for entities."

Self-check

- **Q:** Unequal objects, same hash? **A:** Allowed. **Q:** Equal objects, different hashes? **A:** Broken contract.

TL;DR

Override `equals` and `hashCode` together, on immutable fields.

2.13 Immutability and records

What it is

Immutable objects never change; "changes" return **new** objects. A **record** (Java 16+) declares an immutable data carrier in one line. Analogy: a printed receipt.

The problem it solves

Shared mutable state causes surprise changes and needs locks. Immutable objects are **thread-safe without locks** and safe hash keys.

How it works

Checklist: `final` class (or private constructors + static factories); `private final` fields; no setters; **defensive copies** in and out; don't leak `this`. **Records** generate fields, canonical constructor, accessors (`id()`), `equals`/`hashCode`/`toString`; they're `final`, allow no extra instance fields, may

implement interfaces; a **compact constructor** validates. **Only shallowly immutable**: copy mutable components.

Java example

```java
// File: code/lld/ch02/immutability/Money.java
    private Money(BigDecimal amount, String currency) {
        this.currency = Objects.requireNonNull(currency);
        // UNNECESSARY = refuse (ArithmeticException) anything finer than a cent instead of silently rounding it
        this.amount = amount.setScale(2, RoundingMode.UNNECESSARY);
    }

    public static Money of(String amount, String currency) { return new Money(new BigDecimal(amount), currency); }

    public Money plus(Money other) {
        if (!currency.equals(other.currency)) throw new IllegalArgumentException("currency mismatch");
        return new Money(amount.add(other.amount), currency);   // new object; 'this' is untouched
    }
```

```java
// File: code/lld/ch02/immutability/Order.java
public record Order(String id, List<String> items, Money total) {
    public Order {
        if (id == null || id.isBlank()) throw new IllegalArgumentException("id required");
        items = List.copyOf(items);                // unmodifiable copy: caller's list cannot leak in
    }
    // ... withItem(item, price) returns a modified copy ("wither")
}
```

Full runnable code: code/lld/ch02/immutability/ (see the Full Edition).

Scale fixed at 2 makes `Money.of("5")` equal `Money.of("5.00")`; `items()` is an unmodifiable copy.

Trade-offs

Lock-free safety, no boilerplate; but a new object per change, and records can't hide components or extend classes.

When not to use it

Entities whose state changes (occupied spot, balance) stay **mutable but encapsulated**.

Common mistakes

No defensive copies; thinking `final` freezes the referenced object; records for entities.

Real-world examples

`String`, `BigDecimal`, `java.time`, `List.of`.

In the interview

30-second answer: "Value objects like Money or Ticket are immutable records with compact-constructor validation and defensive copies: thread-safe, safe keys. Entities stay mutable but encapsulated." Records are only shallowly immutable.

Self-check

- **Q:** `final List<String> items` guarantees? **A:** The reference, not the contents.

TL;DR

Final, private final, no setters, defensive copies; records do it (shallowly).

2.14 Access modifiers and packages

What it is

A **package** groups classes (namespace + folder); **access modifiers** set visibility:

Modifier	Class	Package	Subclass elsewhere	Everyone
`private`	✔	✘	✘	✘
(none) package-private	✔	✔	✘	✘
`protected`	✔	✔	✔ (via inheritance)	✘
`public`	✔	✔	✔	✔

Top-level classes: `public` or package-private only.

The problem it solves

Encapsulation becomes **compiler-enforced**; packages hide implementation behind a small API.

How it works

Start `private`; group by **feature**; package-private helpers behind a `public` facade; `protected` only for subclass hooks (it **also** opens the package); Java 9 **modules** export chosen packages.

Java example

```java
// File: code/lld/ch02/access/client/PremiumWallet.java
    public void grantBonus(long cents) {
        bonusCents += cents;                    // OK: protected, accessed through inheritance
        // auditCount++;                        // would NOT compile: package-private in another package
        // balanceCents += cents;               // would NOT compile: private
    }

    @Override public String describe() {       // may WIDEN protected -> public when overriding
        return "[premium] " + super.describe();
    }
```

Full runnable code: code/lld/ch02/access/core/ and code/lld/ch02/access/client/ (see the Full Edition).

`Wallet` (in `core`) has one field per access level; `AuditLog` is a package-private class exposed only through the public `Audits` facade.

Common mistakes

Everything `public`; `protected` fields instead of methods; one package for a whole solution.

In the interview

> **30-second answer:** "Private by default, package-private for feature collaborators, protected only for subclass hooks, public as the deliberate API; packages like model, repository, service, strategy."

Self-check

- **Q:** Non-subclass in another package reads `protected`? **A:** No.

TL;DR

private < package-private < protected < public; default private.

2.15 Generics basics for LLD

What it is

Type-safe code for **any type**: `List<String>`, `Repository<String, User>`; `T`, `ID`, `E` are **type parameters**. Analogy: a "SHOES ONLY" box checked on the way in.

The problem it solves

No `Object` + casts failing at run time: errors become **compile errors**, and one `Repository<ID, T>` serves every entity.

How it works

Generic types and methods; **bounds** (`T extends X`) let you call X's methods; wildcards `? extends T` (producer), `? super T` (consumer): **PECS**; diamond `new Repository<>()`; **erasure**: no `new T()`, no `instanceof List<String>`, no primitives; avoid **raw types**.

Java example

```java
// File: code/lld/ch02/generics/Repository.java
public class Repository<ID, T extends Identifiable<ID>> {
    private final Map<ID, T> store = new LinkedHashMap<>();

    public void save(T item)            { store.put(item.id(), item); }
    public Optional<T> findById(ID id)  { return Optional.ofNullable(store.get(id)); }
    public List<T> findAll()            { return List.copyOf(store.values()); }
}
```

Full runnable code: code/lld/ch02/generics/ (see the Full Edition).

The same class stores `User`s by `String` and `Product`s by `Integer`. `copy(List<? extends T> src, List<? super T> dst)` moves integers into a `List<Number>` (PECS). `ArrayList<String>` and `ArrayList<Integer>` share one runtime class (erasure).

Common mistakes

Raw types; thinking `List<Integer>` is a `List<Number>`; over-generic designs.

Real-world examples

Collections, `Optional<T>`, Spring Data `CrudRepository<T, ID>`.

In the interview

> **30-second answer:** "Generics give type-safe infrastructure like `Repository<ID, T>`; I know bounds, PECS, and erasure."

Self-check

- **Q:** Why no `new T()`? **A:** Erasure; pass a `Supplier<T>`.

TL;DR

Type-safe reuse; bounds, PECS, erasure.

2.16 Coupling and cohesion (the yardsticks)

Cohesion (how well a class's duties belong together) should be **high**; **coupling** (dependence on others' details) should be **low**. Encapsulation and access modifiers cut coupling, single-responsibility raises cohesion, interfaces and composition decouple. Chapters 4–5 name these as SOLID, Law of Demeter, GRASP.

2.17 Object-oriented analysis and design (OOA / OOD / OOP)

What it is

Activity	Question	Talks about
Analysis (OOA)	What is the system, what must it do?	Domain terms: books, members, loans
Design (OOD)	Which objects with which responsibilities?	`Book`, `Member`, `Loan`, who holds whom
Implementation (OOP)	How in code?	`ArrayList`, `HashMap`

The problem it solves

Choosing operations before objects are settled puts them on the wrong class.

How it works: the four steps

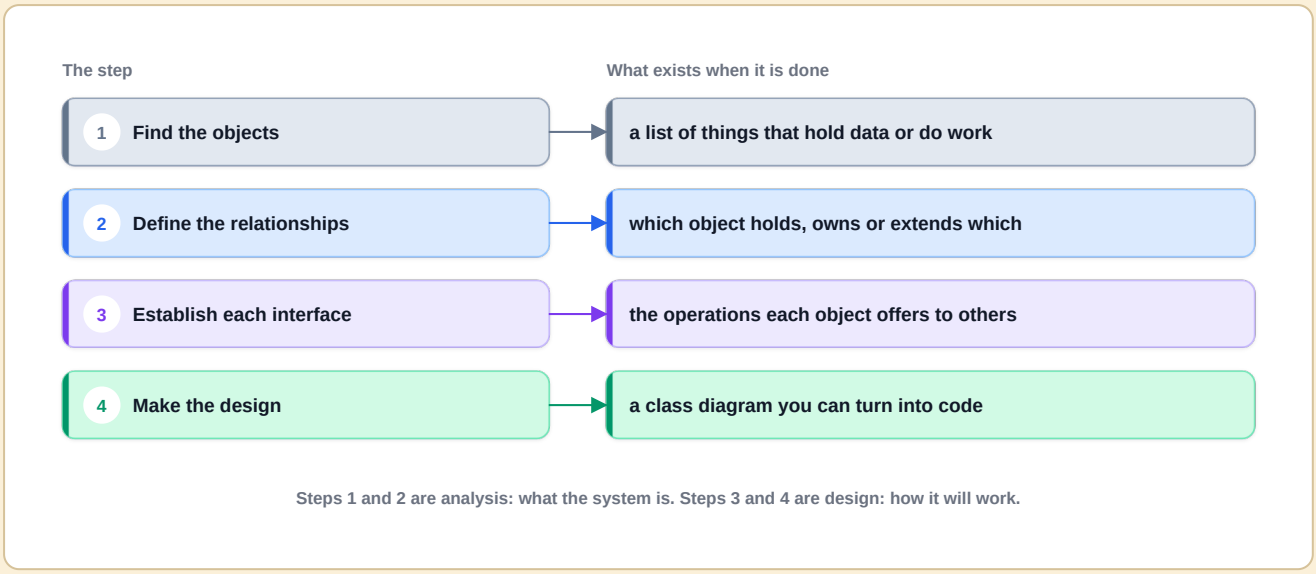


Figure — The four OOAD steps and what exists after each; steps 1–2 are analysis and steps 3–4 are design (course diagram)

1. **Find the objects** (things that store info or do work; single values become fields).
2. **Define relationships** (who references whom, ownership, how many).
3. **Establish each interface** (operations offered to others; the rest is **private**).
4. **Make the design** (one class diagram, checked against the description). Fixed order.

Worked example: "library members borrow books; each book on loan must be returned by a due date"

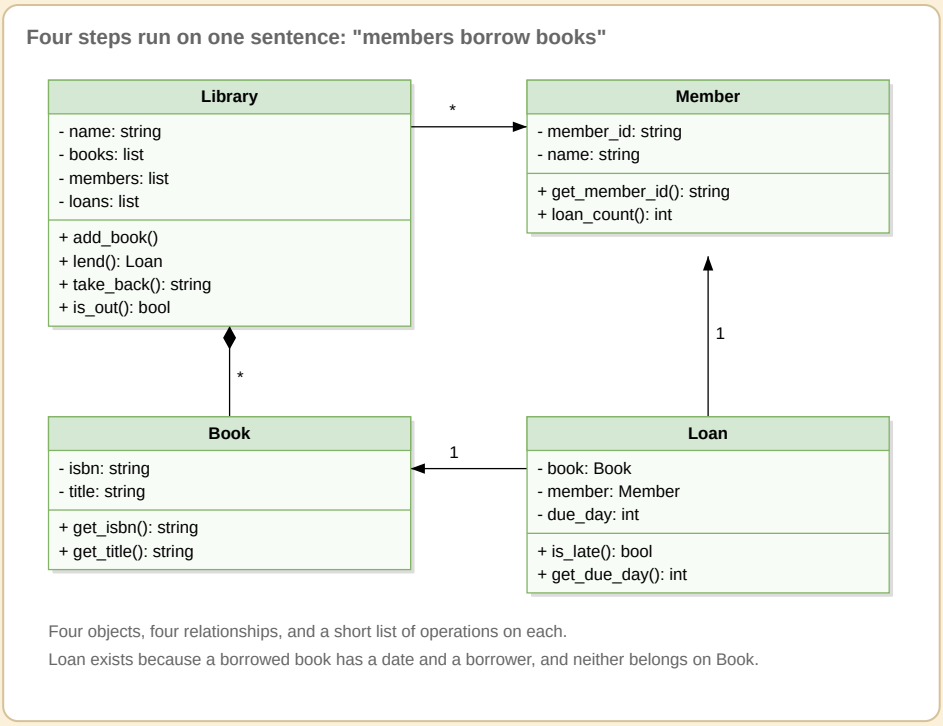
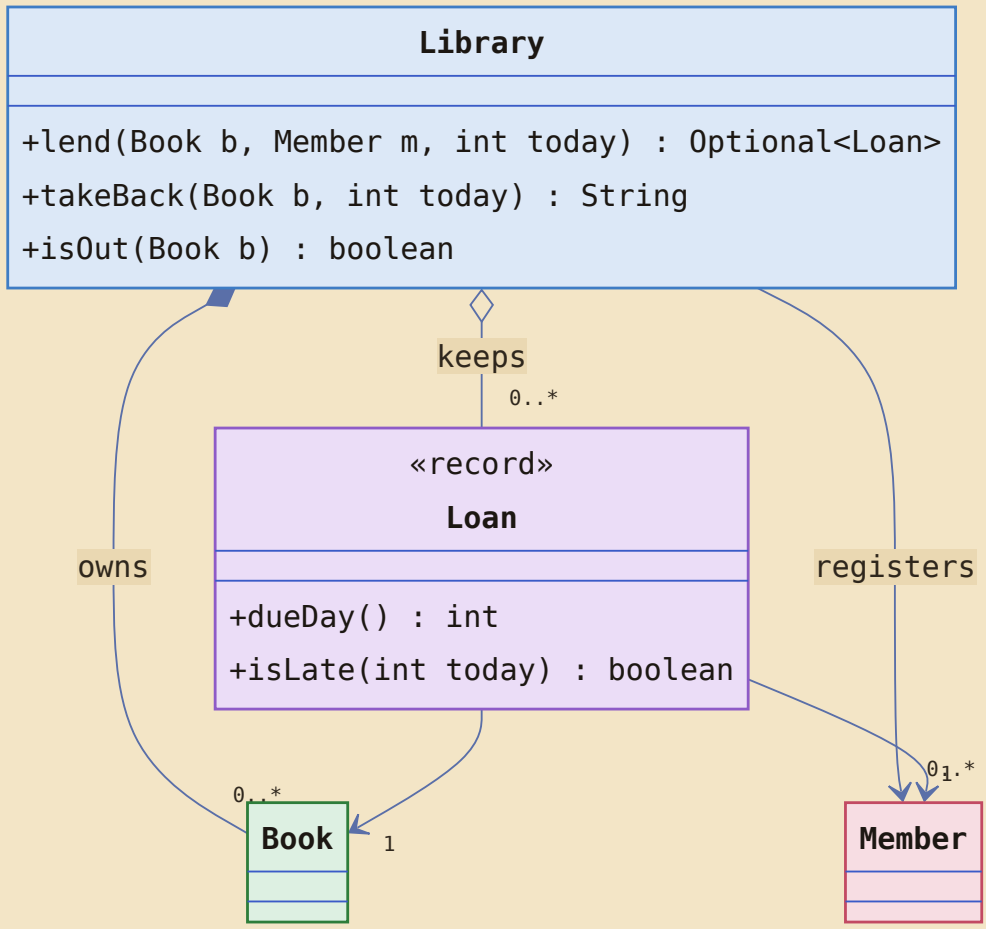


Figure — The four steps applied to one sentence: Library, Member, Book and the connection object Loan (course diagram)

Objects: `Library`, `Member`, `Book`, and **`Loan`** (most often missed); *date* is a field. **Rule: a link with data of its own becomes an object** (`Loan`, `Reservation`, `Ticket`, `Enrollment`). Library **owns** books (composition), **registers** members (association); a loan **points to** one book and one member. Only `Library` can answer "is this book out?" because it keeps the loans.




```java
// File: code/lld/ch02/ooad/Library.java
    public static final int LOAN_DAYS = 14;
    public static final int MAX_LOANS_PER_MEMBER = 5;
    // ...
    public Optional<Loan> lend(Book b, Member m, int today) {
        if (!books.contains(b) || !members.contains(m)) return Optional.empty();
        if (isOut(b) || m.loanCount() >= MAX_LOANS_PER_MEMBER) return Optional.empty();
        Loan loan = new Loan(b, m, today + LOAN_DAYS);
        loans.add(loan);
        m.loanStarted();
        return Optional.of(loan);
    }
```

Full runnable code: code/lld/ch02/ooad/ (see the Full Edition).

Limits became constants; `Loan` is a record with `isLate(today)`; `Member.loanStarted/loanEnded` are package-private so only `Library` changes counts.

Why UML

UML is the standard notation because each symbol records a decision: **filled diamond** = ownership, **plain arrow** = reference, **+ / −** = public/private (Chapter 3).

Three ways this goes wrong

Classes named after tables (`UserRow`) that do nothing; **missing the connection object** (`dueDate` on `Book`); **listing every imaginable operation**.

In the interview

30-second answer: "Analysis first: objects from nouns, relationships with ownership and multiplicity. Then design: each object's public operations (which decides what's private), then

the diagram. A link with its own data becomes a class."

Self-check

- **Q:** Analysis steps? **A:** Find objects, define relationships. **Q:** Parking-lot connection object? **A:** The ticket. **Q:** When is encapsulation decided? **A:** Step 3.

TL;DR

OOA = what; OOD = how with objects; OOP = code. Connections with data become objects.

Chapter summary: in the interview

30-second answer ("pillars of OOP"): "Encapsulation keeps data and its rules in the owning class; abstraction exposes only what callers need; inheritance reuses a parent when is-a truly holds; polymorphism lets one call behave per object, removing type switches. I prefer interfaces and composition, enums for fixed sets, records for values, and consistent equals/hashCode."

- *"Make a class immutable?"* — Final class, private final fields, no setters, defensive copies, or a record.

Self-check (whole chapter)

- **Q:** Dynamic dispatch? **A:** The JVM choosing the override from the actual class at run time.
- **Q:** Where does "which spots fit a truck" live? **A:** On the `VehicleType` enum.
- **Q:** hashCode rule? **A:** Equal objects, equal hash codes.
- **Q:** OOAD steps? **A:** Objects, relationships, interfaces, design.

TL;DR

- OOP = data bundled with the only code allowed to change it.
- Pillars: encapsulation, abstraction, inheritance (true is-a), polymorphism (kills type switches).
- Overloading compile-time; overriding run-time; fields/statics not polymorphic.
- Interfaces = contracts; abstract classes = shared state; enums = fixed sets; composition over inheritance.
- `equals` + `hashCode` together; immutable records for values; encapsulated mutable entities.
- Private by default; generics for type-safe infrastructure; OOA → OOD → OOP.